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HCM·UTE HCMC University of Technology and Engineering

Chương 3
THIẾT KẾ VÀ CÀI ĐẶT HỆ THỐNG NHÚNG



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Hãy thiết kế một hệ thống tự động bật quạt khi nhiệt độ phòng tăng cao

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NỘI DUNG

- Vấn đề thiết kế hệ thống nhúng
- Hardware-Software codesign
- Quá trình thiết kế hệ thống nhúng
- Một số phương pháp luận thiết kế

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EMBEDDED SYSTEM DESIGN ISSUES

- Constraint Issues
- Functional issues
- Real-time issues
- Concurrent issues
- Reactive issues

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EMBEDDED SYSTEM DESIGN ISSUES

- Contraint issues
  - Cost may matter more than speed
  - Long life cycle
  - Reliability/safety
  - Low-power
  - Size/Weight
- Examples: Portable heart-beat monitor
  - Long life cycle (10 years)
  - Reliability (accuracy 99%)
  - Low-power (5 using days)
  - Light weight (<1 kg)

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EMBEDDED SYSTEM DESIGN ISSUES

- Contraint issues
  - Examples for smart home system

No.	Constraints	Note
1	Low price (< 1.000.000 dong)	Correct
2	Ability to detect smoke and fire	Wrong
3	Low power (100mW when idle, 3W when active)	Correct
4	Response time for control < 1ms	?
5	Support remote control by smartphones	?
6	Easy to install	?

Constraints are limitations or restrictions of some parameters of the system

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EMBEDDED SYSTEM DESIGN ISSUES

- Functional issues
  - Safety-critical applications
  - Damage to life, health, economy
  - Affect to environment, society, politics
- Example:
  - Message LED for a shop
    - Display message for customers
    - Malfunctions result in small damage
  - Message LED for a stock market
    - Display stock data
    - Malfunctions could damage to economy
  - Battery charger
    - ?

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EMBEDDED SYSTEM DESIGN ISSUES

- Functional issues
  - Examples for battery charger

No.	Issues	Note
1	Battery can be over-heat, it need to be detected by a sensor	Correct
2	Display a charging current and battery status	Wrong
3	The system must have a fuse for protection of over current	?
4	Support 3 charging modes	?
5	Apply efficient algorithm for fast charging and increase battery life cycle	?

Functional issues are problems which can affect to life, health, economy, environment, society, politics, ethics.

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EMBEDDED SYSTEM DESIGN ISSUES

- Real-time issues
  - Determine whether the system is hard/soft/non real-time
  - Determine the time constraint (delay)
- Example:
  - Door entry alarm
    - Non/soft real-time system: delay < 1 – 2s
  - Video recoder
    - Soft real-time system: delay < 1ms
  - Car airbag system
    - ?
  - Weather temperature monitoring
    - ?

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EMBEDDED SYSTEM DESIGN ISSUES

- Concurrent issues
  - System and environment run concurrently
  - Multi-functions
  - Interface with other systems
  - May need a scheduler to manage concurrent tasks
- Example: Weather temperature monitoring
  - Multi-functions:
    - Read temperature values from the sensor
    - Write data to memtory
    - Display data on LCD

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EMBEDDED SYSTEM DESIGN ISSUES

- Reactive issues
  - Continuous/discontinuous interaction
    - Power on demand
      - Turn ON when using
      - Ex: MP3 player, Tivi system
    - Always ON, once started run forever
      - Continuous interaction with their environment
      - Termination is a bad behavior => watchdog timer
      - Ex: Camera surveillance system, data acquisition system
  - Response to external periodic/non-periodic events
    - Events are periodic: the system needs a scheduler to capture the events
    - Events are non-periodic: the system needs to estimate miss event cases

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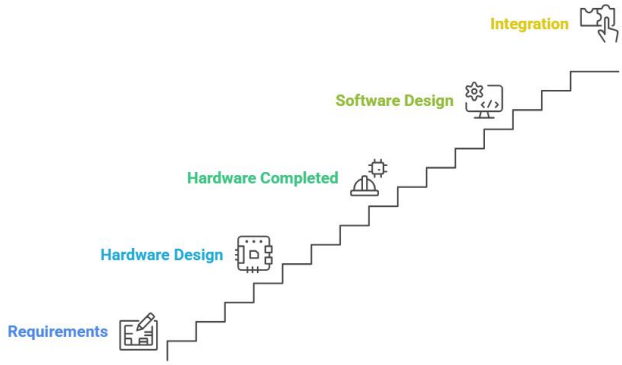
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HARDWARE-SOFTWARE CODESIGN

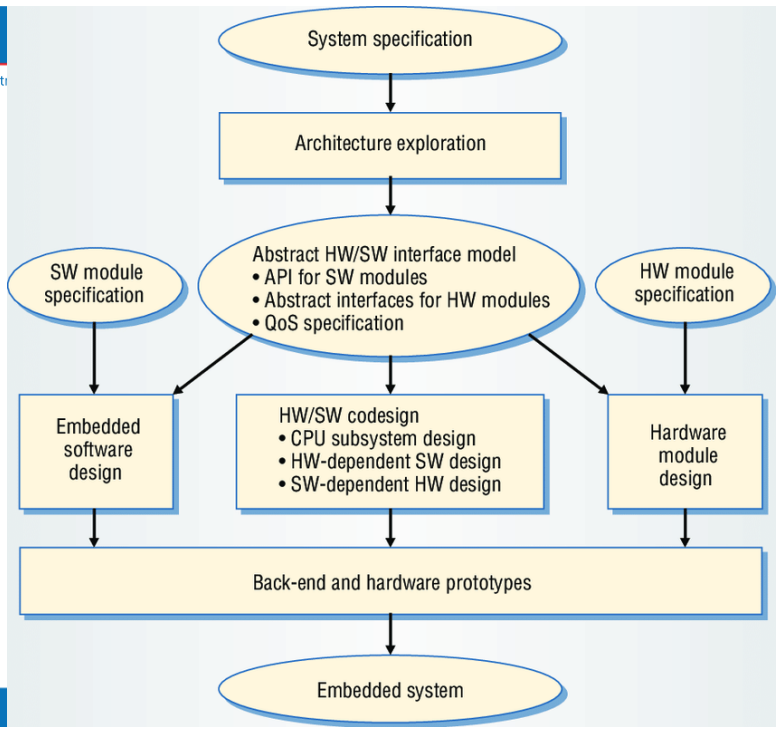
- Hardware–software codesign là phương pháp thiết kế trong đó phần cứng và phần mềm được phát triển song song, có sự trao đổi và điều chỉnh qua lại, thay vì thiết kế phần cứng xong rồi mới lập trình phần mềm một cách tuần tự và tách biệt.
- Vì sao cần HW – SW Codesign:



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HARDWARE-SOFTWARE CODESIGN



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Embedded System Design Process

- Xác định yêu cầu (Requirements)
- Đặc tả yêu cầu (Specification)
- Phân hoạch phần cứng – phần mềm (HW/SW Partitioning)
- Thiết kế giao diện (Interface Design)
- Thiết kế phần cứng (Hardware Design)
- Tích hợp và kiểm tra (Integration & Testing)

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Requirement Defination

A user requirement is a statement that specifies WHAT a product should do, but it does not define HOW it should do it

- WHAT
  - The device shall decrease the temperature of the skin
- HOW
  - The device shall apply moisture to cool down the skin

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Requirement Defination

- May be developed in several ways
  - Talking directly to customers
  - Talking to marketing representatives
  - Providing prototypes to users for comment
- Requirements end up being the specification
  - Containing enough information to begin designing the system architecture

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Requirement Defination

- Functional vs Non-functional
  - Functional requirements
    - Output as a function of input
  - Non-functional requirements
    - Time required to compute output
    - Cost
    - Size, weight, etc.
    - Power consumption
    - reliability

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Requirement Defination

- Validating the requirements
  - Requires understanding what people want and how they communicate it
  - User interface requirements can be refined by using a mock-up
    - May be executed on a PC
  - Physical, nonfunctional models of devices can also help
    - Better idea of size and weight

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Sample Requirements Form

- Name
- Purpose
- Inputs and Outputs
- **Use cases (how user will interact with the system)**
- Functions
- Performance
- Manufacturing Costs
- Power (battery powered? Plugged into a wall?)
- Physical size/weight
- Installation (device is fixed, wall mounted or on desk, etc)
- Certification

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Use Cases

- Describes what the system does
- A collection of operational scenarios that identify the role played by the in response to stimulus, and the output produced by the system
- Initiated by external events that precipitate a system action

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Use Case Composition

- Name
- Description or Sequence of Steps
  - Description of goal to be achieved
  - Describe the sequence of interactions between the system and the actor that transpire from the input system
- Actors
  - Object outside the system that interact with the system (e.g. LCD, Push buttons, sensors,….)
- Assumptions

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Digital Alarm Clock Requirements

- Description:
  - This specification describes and defines the basic requirements of a digital alarm clock. The clock resolution is second, with error rate of 1 sec per day. The block can have 10 alarm time
- External Environment
  - The clock will be used indoor, put on desk or bed side

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Inputs

- 4 small buttons: MENU, UP, DOWN, ENTER
- 1 big button: Stop Alarm

Output

- Display:
  - LCD with back light
  - Date is display as Day/Month/Year
  - Time can be display as am/pm or 24h, up to second
  - The display should be readable in direct sunlight and from wide angle
  - The display will appear similar to this one
  - Power saving mode
  - Day/Night Mode
  - Adaptive Brightness mode
  - Sound: the speaker can generate a sound at the volume of 87 dB, measured by a microphone placed 1 meter away

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Design Specification

- System description
  - Written from a designer point of view
  - System architecture
  - Block diagram if appropriate
- First order functional decomposition
- System behavior
  - Control flow
  - Data flow
  - Signal flow
  - Timing requirements
  - State diagrams
  - Constraints

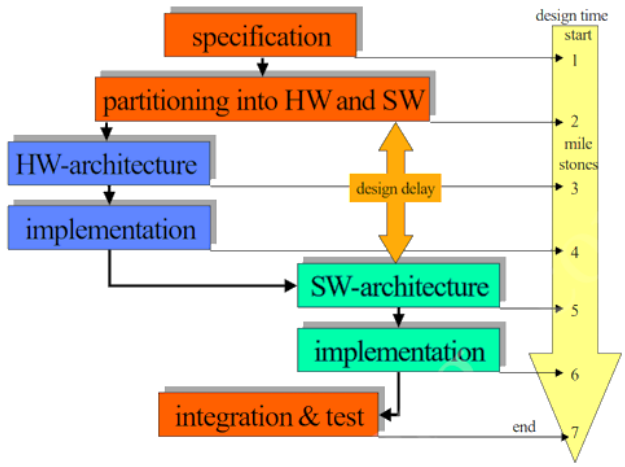
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Design Specification

- Format, structure, and type of the system inputs and outputs
  - Signal level
  - Timing/critical timing
  - Command, Control, Data Formatting
- Memory Maps
- Major modules and their interfaces

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HW/SW Partitioning



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System Partitioning

- System partitioning: divide the system into three parts:
  - HW: microcontroller, memories, peripherals,
  - SW: OS, program, application software
  - Interface: SW driver, HW interface, user interface
- Alternative way: divide the system into HW and SW
  - Which functions should be performed in hardware and which in software?
  - The more functions in software, the lower will be the product cost

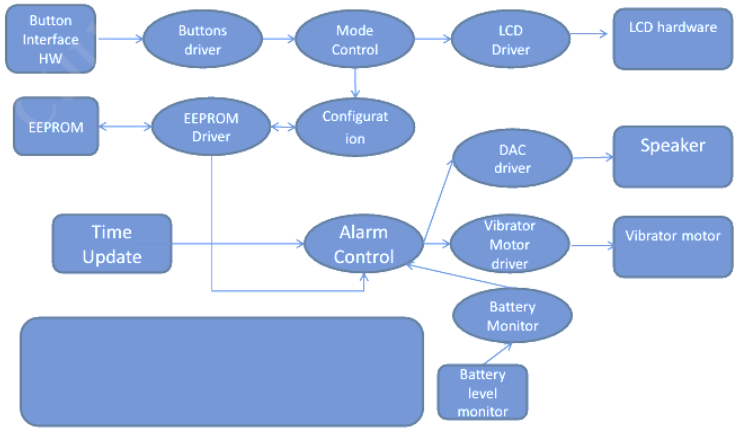
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Design Specification Example

- Description:
  - This specification describes and defines the system level architecture of a digital alarm clock. The clock resolution is second, with error rate of 1 sec per day. The clock can have 10 Alarm time. It supports sound alarm and vibration.
- External Environment:
  - The Clock will be used indoor, put on desk or bed side
  - Temperature range is 10-35 Celcius

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System Block Diagram in detail



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Interface Design

- Button Interface
- Purpose
  - This module provide interface to 5 buttons.
- Requirement
  - Stable, long life button
  - Connector must be tight
  - Buttons are push buttons.

Hardware component	Interface	Component part number	Note
MENU, UP,DOWN, ENTER button	Single end, GPIO, 3.3V/0V		Button is off-board, connect by wires and pin header 2mm pitch
STOPALARM button	Single end, GPIO, 3.3V/0V		Button is off-board, connect by wires and pin header 2mm pitch

Integration & Testing

- Should: testing at each stage

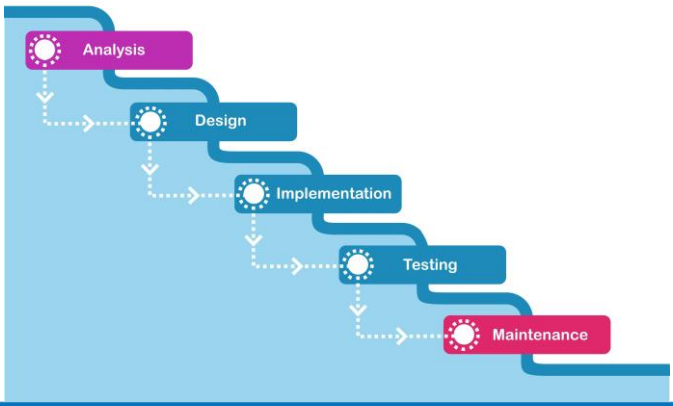
Sensor test
↓
Display test
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Fan test
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Driver test
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Software module test
↓
Integration test
↓
System test

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- Several models can be applied for embedded system design
  - The waterfall model: in which there is a process for developing a system in steps, where results of one step flow into the next step.
  - The spiral model: in which there is a process for developing a system in steps, and throughout the various steps, feedback is obtained and incorporated back into the process.
  - Consecutive refinement model
- Most followed and simple model for building an Embedded System is Waterfall model

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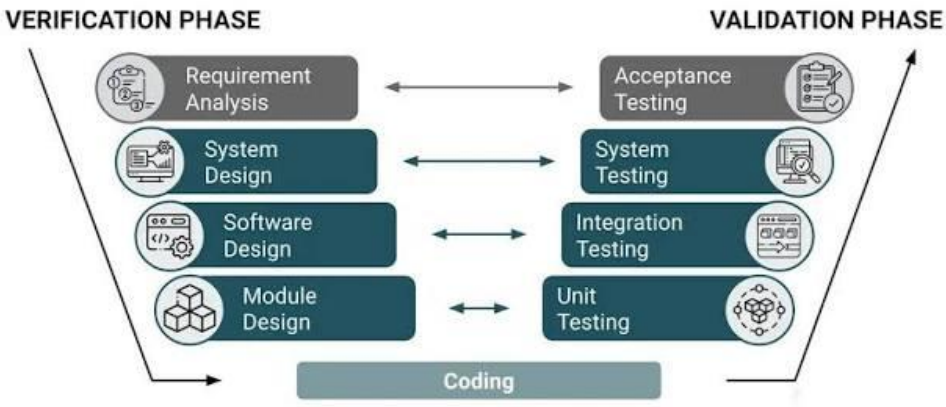
Waterfall



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V Model



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